Day 2 Problem Set

Methods Camp | University of Texas at Austin

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This problem set contains exercises from Day 2 that were originally done in small groups. To reinforce your understanding, please complete these exercises independently.

Exercise 1: Random Variables & Expectation

A survey asks respondents how many times they attended religious services last month. Suppose the distribution is:

Times attended (x)	0	1	2	4
$P(X = x)$	0.4	0.3	0.2	0.1

1.1 Is X discrete or continuous? Why?

1.2 Compute $E[X]$ by hand.

1.3 In your own words, what does $E[X]$ tell us?

1.4 Bonus: if attendance is recoded as $Y = 2X + 1$, what is $E[Y]$? (Use the linearity property.)

Exercise 2: Distributions

2.1 For each scenario below, name the most appropriate distribution (Bernoulli, Binomial, Poisson, or Normal) and briefly justify your choice:

a. Whether a single respondent owns a home (yes/no)

b. The number of home-owners among 200 randomly sampled respondents

c. The number of hate-crime incidents reported in a city per month

d. Adult height in a large population

2.2 For scenario (b), if $p = 0.65$, what is $E[X]$?

Exercise 3: Regression

Suppose a researcher wants to examine the relationship between parents' years of schooling (X) and their adult child's occupational prestige score (Y) across all U.S. families.

3.1 Write the regression equation using index notation. What does the subscript i represent?

3.2 Interpret the slope coefficient.

3.3 Identify the estimand, a plausible estimator, and an example of an estimate.

3.4 Name one plausible factor that could be included in the error term, ε_i.